ELLIPSIS
HEALTH

June 16, 2025

The Honorable Robert F. Kennedy Jr.
Secretary
Department of Health and Human Services
U.S. Department of Health and Human Services
200 Independence Avenue, SW
Washington, D.C. 20201

The Honorable Mehmet Oz, MD, MBA
Administrator
Centers for Medicare and Medicaid Services
7500 Security Blvd
Baltimore, MD 21244

Thomas Keane, MD, MBA
Assistant Secretary for Technology Policy
U.S. Department of Health and Human Services
330 C St SW, Floor 7
Washington, DC 20201

Re: CMS-0042-NC – Health Technology Ecosystem RFI Response Submitted by: Ellipsis Health, Inc.

Dear Secretary Kennedy, Administrator Oz, and Assistant Secretary Keane,

Ellipsis Health ("Ellipsis") appreciates this opportunity to respond to the Health Technology Ecosystem RFI. We share CMS and ASTP/ONC's goal of accelerating responsible technology adoption to empower patients and improve health. Our mission is to alleviate pressing US healthcare challenges: an aging, complex patient population, workforce shortages, fragmented care, and rising chronic disease.

Ellipsis' solution, "Sage," is an empathetic, AI-powered care management agent. Sage extends care team capacity by utilizing multimodal conversation, foundation and large language models (LLMs), millions of real clinical conversations, motivational interviewing techniques, and a proprietary vocal biomarker system for mental health. It enhances preventative care and reduces chronic disease burden by delivering scalable, empathetic, patient-centered care management.

To advance the administration's bold vision to **Make America Healthy Again**, we recommend that CMS and ASTP/ONC implement a policy and regulatory agenda that enables AI-powered care management solutions that help providers and health plans enhance prevention, improve patient care, manage chronic diseases, and empower individuals toward self-care. Our comments focus on sections B (Patients and Caregivers), C (Providers), E (Technology Vendors, Data Providers, and Networks), and F (Value-Based Organizations) of the RFI.

1. Our Vision for a Healthier America with AI Care Management

The US healthcare system faces critical challenges: a fragmented patient journey, escalating chronic disease burden, and severe clinician shortages leading to burnout. This reactive system primarily treats illness, rather than preventing it. Ellipsis Health envisions a proactive, preventative healthcare future empowered by AI. Our AI care manager, Sage, delivers scalable, empathetic, and personalized support, transforming care delivery and driving prevention. Sage addresses current systemic weaknesses by:

- **Automating administrative tasks:** Freeing clinicians for higher-value patient care. **(PR-2, PR-4)**
- **Expanding access to high-quality care:** Reaching patients 24/7, across diverse languages, and even in areas with limited digital access. **(PC-6, PR-1)**
- **Facilitating personalized prevention and self-management:** Empowering individuals to manage their health proactively. **(PC-1, PC-5)**
- **Integrating behavioral health:** Proactively identifying mental health needs through advanced vocal biomarker technology. **(VB-2, VB-3)**
- **Enhancing interoperability:** Enabling seamless data exchange to unify the patient's health record. **(TD-4, TD-13, TD-15)**
- **Driving Value-Based Care success:** Improving outcomes and closing care gaps efficiently. **(VB-1, VB-3)**

We urge CMS to lead in fostering this innovation through supportive policies, pilots, and collaborations. By embracing responsible AI, we can improve health outcomes and ensure every Medicare & Medicaid beneficiary receives the support needed to make informed health decisions and achieve their health goals.

Demo patient, Lee Wang engages with Ellipsis Health's care management agent, "Sage" to help improve adherence to her medications. Sage is able to empathetically guide Lee by listening to her concerns, identifying barriers and providing recommendations for solutions that enable Lee to better care for herself. The link for this demo is available here: https://vimeo.com/1093869046/60218ad3c0

2. Support Providers and Payors in Reducing Clinical and Administrative Burdens (PR)

Challenge: Today's healthcare system is characterized by fragmented patient journeys, limited access to primary and specialty care, and obstacles to promoting healthy lifestyles. Economic

realities and clinician shortages lead to thin margins and staff burnout, exacerbated by time-consuming administrative tasks and disparate data systems.

Ellipsis Solution: Sage Streamlines Operations & Extends Care Capacity Ellipsis Health's Sage acts as an AI care manager, coalescing disparate information to provide a seamless, integrated care coordination experience. Sage streamlines healthcare operations by automating routine administrative tasks such as appointment scheduling, reminders, medical record management, billing, and preliminary health assessments. This significantly reduces the workload on human staff, allowing clinicians to practice at the top of their license. Sage operates 24/7, checking physical and behavioral health, discussing medication, managing symptoms, providing next steps, and documenting interactions. In emergencies, Sage can escalate calls to live nurses or emergency lines. **(PR-2; PC-1).**

Sage can handle a myriad of diverse tasks seamlessly, including verifying identity, introducing the care management program, checking insurance and employer eligibility, as well as programmatic appropriateness, capturing recent labs and meds, checking the ability to wear a fitness tracker, scheduling a lab appointment, and scheduling onboarding visits all while keeping the patient engaged. In an internal study conducted in partnership with a specialty care management company, Sage was able to complete a typical patient onboarding workflow taking an average member outreach specialist six weeks to fill a clinical staff caseload, in one week given Sage's ability to simultaneously conduct almost unlimited inbound and outbound calls to enroll members. Sage also was able to complete all onboarding tasks during an enrollment call at a rate that was 39% higher than the average teammate. Lastly, Sage was able to complete clinical intake calls as part of this study in less than 20 minutes on average compared to an average call taking 32 minutes before the partnership between Ellipsis Health and the specialty care management company. Public studies confirm similar outcomes - one study indicated that AI chatbots can cut consultation wait times by 15% and handle up to 80% of routine queries, with one orthopedic practice saving over 95 hours of phone time by offloading inquiries to an AI chatbot.[1] This efficiency is critical for financially and operationally strained environments. **(PR-2).**

Recommended Actions to Reduce Burden and Foster AI Adoption:

- **Establish AI-Focused Reimbursement Codes:** CMS should create new reimbursement codes or modify existing ones (e.g., Chronic Care Management (CCM), Transitional Care Management (TCM), Principal Care Management (PCM), Remote Patient Monitoring (RPM) to specifically cover AI-enhanced care management services. This ensures the financial viability of providers adopting these solutions **(PR-1a; PR-4).**
- **Mitigate Regulatory Burdens for Medicare Advantage (MA) Plans:** CMS should clarify and mitigate any regulatory obstacles that prevent MA plans from reimbursing AI care management services, encouraging broader adoption. **(PR-1a).**
- **Develop Medicaid Guidance:** CMS should issue comprehensive guidance to state Medicaid agencies on how to best leverage, reimburse, and ensure the safety of AI care management solutions, supporting access for low-income beneficiaries. **(PR-1a).**

[1] Miliard, M. (2024, April 4). AI-powered patient communication tools help one orthopedic practice save $405,000. Healthcare IT News.

- **Evaluate Direct Contracting Opportunities:** CMS, in coordination with agencies like the Veterans Administration (VA), should evaluate direct contracting with AI care managers to support care for specific populations, leveraging government scale for efficient procurement. **(PR-1a).**
- **Fund Pilot Programs for Administrative APIs:** CMS could stimulate integration by funding pilot programs for EHR vendors and providers. These programs should focus on developing and testing **ready-to-use** API modules for common administrative workflows (e.g., appointments, forms), specifically to support digital health apps, including AI agents. **(PR-2; PR-4).**

3. Expanding Access & Empowering Patients through Personalized Prevention (PC)

Challenge: Healthcare access remains uneven, particularly for high-need groups and those in rural areas. Many Medicare beneficiaries lack experience with digital tools, and existing solutions often fail to meet patients where they are, hindering engagement with preventative care. Chronic diseases pose a significant health burden in the U.S., affecting 60% of Americans with at least one condition and 40% with multiple conditions, as reported by the CDC. These conditions, encompassing heart disease, cancer, diabetes, and obesity, are major contributors to mortality and morbidity. **(PC-1; PC-6; PC-8).**

Ellipsis Solution: Sage Delivers Accessible, Empathic, Preventative Support *Sage* extends the reach, scale, and scope of healthcare service providers and delivers a more patient-centric, empathetic experience that meets patients where they are. Sage fundamentally alters patient access and responsiveness. Functioning 24/7 with rapid response times, Sage interacts with patients via traditional phone calls, making it accessible even for those without broadband or digital literacy. It provides support in common US languages (English, Spanish, and more) and integrates regional dialects for maximum engagement. Sage's empathetic, multimodal conversational capabilities foster trust, making patients comfortable discussing their health. **(PC-6; PR-1).**

Sage focuses on preventative care and patient self-management. It leverages key patient data, including medical history, symptoms, and lifestyle information, to provide personalized health navigation and chronic care management services. This enables tailored advice and interventions, critical for effective chronic disease management and promotion of healthy behaviors. Sage actively encourages behavioral modifications, supports self-management of diverse health conditions, and offers ongoing guidance, such as promoting adherence to dietary, exercise, or medication regimens through targeted behavioral nudges. Sage also helps patients by surfacing relevant healthcare data, scheduling medical and lab appointments, and translating health insurance benefits into easy-to-understand options for services.

As the Wall Street Journal recently noted, Sage is tailored for patients with specific diseases such as diabetes and cancer. The capability of *Sage* to deliver personalized care and behavioral prompts empowers patients in their health journey, fostering a patient-centric model wherein individuals receive consistent support for the proactive management of their own health. This

continuous support can lead to more sustained improvements in health outcomes compared to episodic interventions. For example, a study indicates that 61% of patients reported AI chatbots aided symptom self-management, and patients found conversational AI engaging for self-management support.[2] **(PC-1; PC-8).**

Studies confirm AI care managers improve patient engagement (up to 30% increase) and care plan adherence (as high as 97%)[3]. Patients report enhanced clarity. A study found that patients reported greater clarity of information (3.73 vs. 3.62 out of 4, $p<0.05$) and overall satisfaction (4.58 vs. 4.42 out of 5, $p < 0.05$) with AI-assisted conversations compared to standard care, while maintaining equivalent levels of trust and perceived empathy.[4] Furthermore, AI-based healthcare chatbots enhance condition management, promote healthy behaviors, and support mental health, increasing medication adherence by 15%.[5] Sage's continuous support leads to more sustained health improvements and can significantly reduce hospital readmissions (by up to 25% in some studies)[6], shifting care from reactive to proactive. **(PC-1; PC-5).**

Recommended Actions to Expand Access and Empower Patients:

- **Accelerate Coverage Pathways for AI Tools:** CMS should establish an expedited Medicare National Coverage Determination (NCD) process for certified AI digital health tools. This accelerated process, guided by AI technical and clinical specialists, patients, and caregivers, must allow for ongoing product refinements without burdensome re-evaluations. A framework similar to the Transitional Coverage of Emerging Technologies (TCET) for FDA-approved breakthrough devices would be highly advantageous. **(PC-5a; TD-1).**
- **Launch Nationwide Awareness and Education Initiatives:** Implement programs to raise awareness and provide education on credible digital health solutions for Medicare and Medicaid beneficiaries and caregivers, focusing on rural and underserved areas. This will build trust and encourage adoption. **(PC-5).**
- **Incentivize Preventative AI:** Explore incentives or bonus programs within existing CMS models (e.g., Medicare Shared Savings Program, CMMI models) for providers and ACOs that effectively deploy AI-driven preventative outreach and self-management tools, linking their use to improved health outcomes and cost savings. **(VB-1).**
- **Support a Unique Patient Identifier:** CMS should support the creation and adoption of a universal patient identifier. A UPI would be transformative by providing everyone with

[2] Wang, F., Casalino, L. P., & Khullar, D. (2023). *Deep learning in medicine—promise, progress, and challenges*. New England Journal of Medicine, 388(11), 995–997
[3] Khan, M. A. A., Alam, M. K., & Mohd Salleh, N. A. S. (2024). *Revolutionizing medicine: Chatbots as catalysts for improved diagnosis, treatment, and patient support*. Cureus.
[4] Xie, S., Mehta, H., Yang, X., Khattar, D., Gao, W., Kalidindi, A., ... & Liang, P. (2024). *AgentTuning: Enhancing LLMs as Agents* [Preprint]. arXiv.
[5] Sathian, B., Asim, M., Banerjee, I., Pizarro, A. B., Mekkodathil, A., van Teijlingen, E., & Simkhada, P. (2023). *Artificial Intelligence (AI) applications for COVID-19 pandemic*. Diabetes & Metabolic Syndrome: Clinical Research & Reviews, 17(2), 102635.
[6] Soni, S., Sharma, A., & Gaba, A. (2024). *Artificial intelligence in healthcare: Friend or foe?* Cureus, 16(4), e56347.

a single, privacy-protected token recognized across all healthcare touchpoints, enabling reliable patient matching and consistent clinical context for AI agents. **(PC-14).**

- **Support Pilot Programs for Preventative Outreach:** CMS could support pilot programs specifically where AI-driven preventative outreach is deployed at scale for chronic disease management, prevention, and screenings. These pilots should support care for a diverse set of beneficiaries and include publicly available evaluations, providing real-world evidence and encouraging adoption of effective tools. **(PC-5a).**

4. Accelerating Interoperability (TD)

Challenge: The US healthcare system's historical fragmentation and data siloing hinder quality, access, and cost efficiency. Proprietary legacy systems create barriers to AI integration and patients' ability to access their own health data. The "last mile" implementation problem, including mismatched data standards, impedes the realization of AI's benefits. **(TD-7; TD-13; TD-19).**

Ellipsis Solution: Sage Leverages Interoperability for Coordinated Care. While Sage does not require comprehensive EHR data to be effective (only basic demographics are necessary), it generates structured data that can be seamlessly shared with providers and integrated into EHRs. Sage's efficacy is enhanced by accessing a broader range of patient data for care navigation and management. **(TD-4; PR-3).**

Ellipsis advocates for regulatory standards that mandate EHR manufacturers and payers to adopt FHIR-based APIs and to advance bulk FHIR API capabilities, enabling efficient sharing of a patient's comprehensive health record. This will improve Sage's ability to integrate into EHRs, access more patient data, and provide personalized services. **(TD-10; TD-15; PA-5).**

Recommended Actions to Accelerate Interoperability:

- **Enforce and Expand FHIR-Based API Mandates:** CMS and ONC must ensure the timely implementation and rigorous enforcement of existing regulations requiring FHIR-based API adoption. Build on these regulations to require widespread FHIR API adoption across all sectors of the healthcare system (e.g., laboratories, imaging centers, pharmacies). **(TD-4; PA-2).**
- **Advance Full Electronic Health Information (EHI) Exchange:** Accelerate efforts to make all aspects of a patient's health information interoperable, including clinical notes, free text, health insurance information, emergency contacts, social determinants of health (SDOH), problem and medication lists, laboratory and diagnostic results, care events and encounters, and upcoming and missed appointments. This includes supporting bulk FHIR APIs for cohort-based patient outreach. **(TD-13; TD-15).**
- **Leverage EHR Certification for Standards Adoption:** Continue to leverage the ONC Health IT Certification Program to drive the adoption of USCDI and FHIR standards, dismantling barriers erected by legacy EHR manufacturers. This should include tightening certification criteria to ensure APIs cover "all aspects of the patient's chart,"

including unstructured data, possibly through emerging text or image exchange standards. **(TD-2, TD-7; TD-8; TD-9, PC-9).**

- **Support a Nationwide Provider Directory and TEFCA:** Establish a standardized, nationwide provider directory of FHIR endpoints to streamline connections between systems. Continue building on existing efforts to implement the Trusted Exchange Framework and Common Agreement (TEFCA) to address longstanding interoperability and data exchange challenges, and provide support to independent healthcare providers for connecting to HIEs and TEFCA. **(TD-5; PC-10; PR-6).**
- **Enhance Healthcare Price Transparency:** Continue building on existing efforts to advance healthcare price transparency to support digital patient navigation solutions like Sage, enabling patients to make informed, affordable healthcare choices. **(TD-19).**
- **Broaden Data-Sharing Mandates for APM Entities:** HHS should consider broadening data-sharing mandates to cover APM entities. For example, when a beneficiary in an ACO is admitted to any hospital, ensure the ACO's care team is notified (perhaps via the QHIN/TEFCA network or other intermediaries). This would empower AI agents or other tools to react promptly, such as calling the patient post-discharge, regardless of hospitalization location. **(VB-8; PR-7).**

5. Integrating Behavioral Health and Driving Value-Based Care Success (VB)

Challenge: The US faces a significant behavioral health crisis, with mental health inequities contributing substantially to avoidable healthcare expenditures, projected to reach $1.3 trillion by 2040.[7] Workforce shortages impede access to behavioral health services, and co-morbid mental/physical conditions amplify healthcare costs. Value-Based Care (VBC) organizations often struggle with care gap closure and leveraging new technologies effectively to manage populations and outcomes. **(VB-2; VB-3; VB-4).**

Ellipsis Solution: Sage Proactively Addresses Behavioral Health & Optimizes VBC Ellipsis has developed a proprietary **vocal biomarker** integrated into Sage's patient interactions. This technology detects the severity of depression and anxiety symptoms through semantic and acoustic analysis of spoken language, proactively identifying mental health needs. In a research study in press at JMIR AI (included as an attachment), Ellipsis' vocal biomarker demonstrated high accuracy in predicting depression severity across more than 2,086 recordings of case management calls with verbally administered PHQ-9 surveys of Highmark Blue Cross Blue Shield members. As part of the study, the recordings were divided into a Development set (n=1336) and Blind set (n=671) and PHQ-8 scores were provided for the Development set for vocal biomarker refinement while PHQ-8 scores from the Blind set were withheld until after the vocal biomarker model severity output was reported. The Concordance Correlation Coefficient (CCC) for the test of the vocal biomarker model on the Development set was 0.57 and for the Blind set 0.54, while the Mean Absolute Error (MAE) for the Development set was 3.91 and for the Blind set was 4.06, demonstrating strong model performance. Overall, the findings suggest

[7] Kamath, N., Xu, H., Ramaswamy, R., & Holliday, D. (2022, October 4). *The economic burden of mental health inequities in the United States*. Deloitte Insights.

that speech and Ellipsis' vocal biomarker model may have significant potential for detection and measurement of depression severity categories that may enhance treatment, improve clinical decision-making, and enable truly personalized treatment recommendations. An initial FDA application for the vocal biomarker technology is pending. Public studies, likewise, confirm the effectiveness of AI in behavioral health; one such study identified that users found mental health chatbots helpful and comfortable for screening and referral, with 83% of screened individuals receiving a referral and 50% of those referred subsequently scheduling an appointment for care.[8] **(VB-2; VB-3).**

For Value-Based Care, Sage provides a scalable, cost-effective solution for patient-centered, longitudinal care management crucial for reducing unnecessary emergency department (ED) visits and inpatient admissions. Sage's capabilities include:

- **Data Aggregation:** This utilizes AI to consolidate diverse datasets (claims, clinical, demographic, behavioral, socioeconomic, patient-reported) into a readily consumable format for providers.
- **Dynamic Care Planning:** Enhances the frequency and quality of patient interactions, generating concise summaries and recommending actions based on clinical protocols, thereby augmenting the care team.
- **Care Gap Closure:** Scales patient outreach and facilitates empathetic interactions to systematically address care gaps, leading to enhanced performance on provider quality measures, as well as Stars and HEDIS metrics. **(VB-3; VB-11).**

Recommended Actions to Integrate Behavioral Health and Incentivize VBC Success:

- **Reimbursement for AI-Enabled Behavioral Health:** CMS should develop payment codes for behavioral health-related screening, diagnosis, and general integration services provided by AI care managers, including those using vocal biomarker technology. These codes should be billable alongside existing care management and remote patient monitoring codes. **(VB-13).**
- **Incentivize Vocal Biomarkers in Value-Based Care:** Explore how incentives for vocal biomarker technology can support the success of value-based care initiatives, and examine scalable value-based payment mechanisms for this technology. **(VB-13).**
- **Expand Contracting Opportunities for Digital AI Tools:** Expand contracting opportunities for digital AI tools to support participants across CMS value-based care programs. Standardize and implement incentives for digital health adoption across CMMI models. **(VB-1; VB-7).**
- **Pilot AI in Value-Based Care Models:** CMMI should pilot incentives for Accountable Care Organizations (ACOs) and other Alternative Payment Models (APMs) to leverage AI care managers, potentially designating preferred vendors or directly procuring services to leverage CMS's scale. CMS could also facilitate bulk procurement or deployment of proven AI solutions for smaller ACOs who might not otherwise afford them, similar to how CMS has offered subsidized tools in the past. **(VB-1).**

[8] Haider, Z., & Jalal, H. (2023). *Evaluating the effectiveness of digital mental health interventions: A meta-review*. Cureus, 15(3), e35949.

- **Promote Coordination of Physical and Mental Health:** Develop policy to explicitly support and incentivize the adoption of effective biomarker technologies, particularly among participants in value-based care models, to advance primary care-behavioral health integration. Coordinate regulations across health care agencies (CMS, ONC/ASTP, FDA) to facilitate access to proven biomarker technologies. **(VB-2; VB-8).**
- **Allow Substitution of CEHRT Functions with Certified AI:** CMS could allow substituting certain CEHRT functions with certified AI capabilities. If an APM can demonstrate that a certified AI agent performs a function (e.g., patient education, care planning documentation) as effectively as a CEHRT module, that should count toward meeting technology use requirements, creating an alternative compliance pathway for innovative non-EHR solutions. **(VB-8).**
- **Provide Performance Evaluation Credit for Digital Health Engagement:** CMS should consider incentivizing the adoption of digital health coaching among ACOs and other APM participants and subsequently evaluate how digital health coaching impacts patient outcomes and programmatic spending in MSSP and other programs. For example, CMS could provide a bonus for ACOs which have a higher percentage of their patients engaging with digital health coaching and more closely evaluate how this engagement impacts quality scores and total cost of care. **(VB-1).**

6. Conclusion

Ellipsis Health applauds CMS's goal of accelerating the adoption of safe and proven health technologies. The substantial increase in healthcare-focused AI adoption, with over 70% of healthcare organizations incorporating them and market valuations anticipated to reach $10.26 billion by 2034, underscores a compelling value proposition driven by systemic pressures: rising chronic diseases, an aging demographic, and healthcare workforce shortages. AI care management models, such as Sage, provide scalable digital solutions to mitigate these pressures, optimizing healthcare professionals' time allocation and alleviating burnout. This enhances quality and is essential to sustain and elevate care delivery.

Ellipsis Health strongly encourages CMS to assume leadership in facilitating trustworthy, empathetic AI care management agents capable of enhancing prevention and mitigating the burden of chronic disease. Sage is today demonstrating positive outcomes for Medicare beneficiaries and their care teams. Through clear regulatory support, targeted incentives, and accelerated interoperability, CMS can catalyze the future of AI-powered chronic care management that improves quality and patient experience while reducing costs.

Sincerely,

Mainul I. Mondal

CEO/Founder, Ellipsis Health

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